

| 1 Chapter 1: An Introduction to Veridical Data Science

| 1.1 The PCS Framework

- **Predictability**: Your conclusions are considered to be predictable if you can show that they reemerge in relevant future data and/or are in line with existing domain knowledge.
- **Computability**: You typically don't need to demonstrate that your results are computable; rather you will use computation to conduct and evaluate your analyses, as well as to create data-inspired simulations.
- **Stability**: Your conclusions are considered stable if you can show that they remain fairly consistent when realistic perturbations (changes/modifications) are made to the underlying data, judgment calls, and algorithmic choices.

| 1.2 The Training, Validation, and Test Datasets

- **training set**: at least 50 percent of the original data; used to conduct your explorations, analyses, and to train your algorithms
- **validation set**: typically use 20 percent to 40 percent of the original data; used to demonstrate that your results reemerge
- **test set**: typically 20 percent of the original data; used to provide a final independent assessment of an algorithm

| 1.3 Dataset Splitting

- **Time-based split**: If your data was collected over time and you will be applying an algorithm that you have trained to future data
- **Group-based split**: If your data has a natural grouping, and you will be applying an algorithm that you train to a new group of data points
- **Random split**: If the current data points are all more or less exchangeable with one another

| 1.4 Investigate the Uncertainty

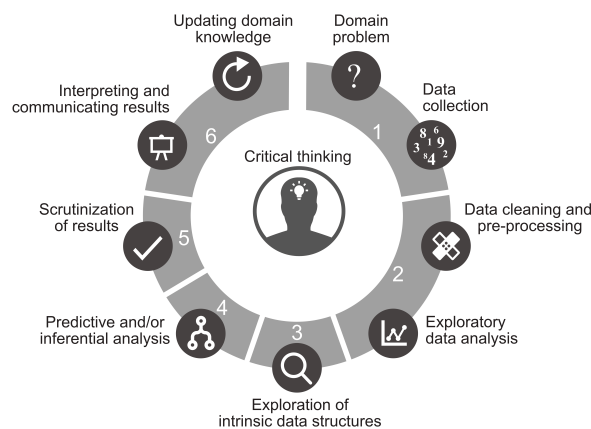
- recompute the results using slightly **different perturbed versions of the data**
- compare the results that are computed using **alternative versions of the data that have been cleaned and preprocessed**
- evaluate the results using **future data**

| 2 Chapter 2: The Data Science Life Cycle

| 2.1 The Data Science Life Cycle

1. Domain problem formulation and data collection
 - Formulating a Question
 - Data Collection
 - Making a Plan for Evaluating Predictability
2. Data cleaning, preprocessing, and exploratory data analysis
 - Data Cleaning
 - Preprocessing
 - Exploratory Data Analysis
3. Exploration of intrinsic data structures (dimensionality reduction and clustering)
 - Dimensionality reduction analysis
 - Cluster analysis
4. Predictive and/or inferential analysis
 - Prediction
 - Data-Driven Inference

5. Evaluation of results
6. Communication of results



3 Chapter 3: Setting Up Your Data Science Project

3.1 An Example File Structure (Using R)

```
1 data/
2   data_documentation/
3     codebook.txt
4   data.csv
5 dslc_documentation/
6   01_cleaning.qmd
7   02_eda.qmd
8   03_clustering.qmd
9   04_prediction.qmd
10  functions/
11    cleanData.R
12    preProcessData.R
13  README.md
```

- **data/**
 - raw dataset
 - data documentation information
- **dslc_documentation/**
 - separate **.qmd** Quarto file (for R) or **.ipynb** Jupyter Notebook (for Python) for conducting and documenting the code-based explorations and analysis at each stage of the DSLC (with numeric prefix)
 - **functions/** subfolder with the **.R** (R) or **.py** (Python) scripts
- **README.md**

3.2 Tips and Tricks for Reproducibility

- Avoid handling raw data or doing manual modifications
- Regularly rerun your code from start to finish in a fresh R/Python session
- Tidy up your code files
- Use version control (e.g., Git)
- Document everything
- Use up-to-date software

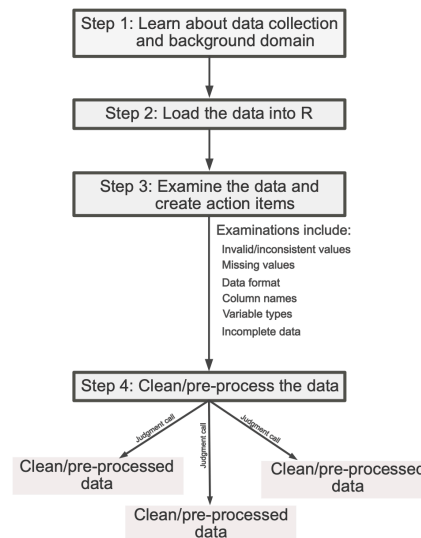
4 Chapter 4: Data Preparation

4.1 Overall Process of Data Preparation

1. **cleaning** your data by modifying it so it is appropriately formatted and unambiguous
2. **preprocessing** your data so it conforms to the formatting requirements required by whatever computational algorithms you plan to use to answer your question

4.2 Four-Step Data Cleaning Procedure

- Step 1: Learn about the data collection process and the problem domain
- Step 2: Load the data
- Step 3: Examine the data and create action items
- Step 4: Clean the data



4.2.1 Step 1: Learn About the Data Collection Process and the Problem Domain

Learning About the Domain Area

- What does each variable measure?
- How was the data collected?
- What are the observational units?
- Is the data relevant to my project?
- What questions do I have, and what assumptions am I making?

4.2.2 Step 2: Load the Data

Identify How to Load Data:

- How your data is being stored?
- If your data contains multiple tables, is there a key variable that connects them?
- What parts of the data are relevant to the question being asked?

Checking That Data Has Been Loaded Correctly:

- Look at the first and last 10 rows of your loaded data and compare it to the first 10 rows of the raw data file to confirm that they match
- Check that the dimension of your data matches what you expect

4.2.3 Step 3: Examine the Data and Create Action Items

Messy Data Trait 1: Invalid or Inconsistent Values

Suggested Examination:

- Look at randomly selected rows in the data, as well as sequential rows
- Print out the **minimum, maximum, and average values** of each numeric column
- Look at **histograms** of the distributions of each numeric variable

- Print out the **unique values** of categorical variables

Possible Action Items:

- Leave them as they are
- Replace the invalid/inconsistent values with the “correct” value
- Replace invalid values with missing values
- Convert measurements with different units to a common unit

Messy Data Trait 2: Improperly Formatted Missing Values

Suggested Examination:

- Print out the **minimum and maximum** of each numeric column
- Print out the **unique values** of categorical variables
- Look at **histograms** of the distributions of each numeric variable
- Print the **number (and the proportion) of missing values** in each column
- **Visualize the patterns of missing data** (e.g., using a heatmap)

Possible Action Items:

- Replace ambiguous missing values (e.g., 99999 or "missing") with explicit missing values (NA)
- Replace them with "right" values and justify your choice

⚠ Remark: A Note on Imputing Missing Values ▾

Unless it is absolutely necessary (or your data contains a lot of observations—at least in the tens of thousands), we don't recommend imputing variables that have more than half of their values missing.

Messy Data Trait 3: Nonstandard Data Format

Suggested Examination:

- What are the observational units? Is there a single row for each observational unit?
- Does each column correspond to a single type of measurement?

Possible Action Items:

- Pivot your data to a wider format if it is in a long format, or to a longer format if it is in an overly wide format
- Aggregate multiple measurements for individual observational units
- Separate columns that contain merged measurements

Messy Data Trait 4: Messy Column Names

Suggested Examination:

- Simply print the column names

Possible Action Items:

Modify column names to follow the style guide:

- lowercase
- human-readable
- words should be separated by underscores

Messy Data Trait 5: Improper Variable Types

Suggested Examination:

- Print the type of each column
- Look at individual values in each column

Possible Action Items:

- Converting each variable to the most appropriate type
- Reformatting the levels of categorical variables
- Aggregating uncommon/rare levels of a categorical variable

Messy Data Trait 6: Incomplete Data

Suggested Examination:

- Check that every observational unit appears in the data **exactly once**
- Check that the number of rows in the data matches what you expect, such as from the data documentation

Possible Action Items:

- If any observational units are missing from the data, add them to the data and populate unknown entries with missing values

| 4.2.4 Step 4: Clean the Data

Do not recommend saving a copy of your clean datasets to load for your analysis

| 4.2.5 Additional Common Preprocessing Steps

- Standardizing or normalizing variables that are on different scales
- Transforming variables so they follow a more Gaussian
- Splitting categorical variables into binary variables, called “one-hot encoding” in ML
- Featurization